

2(a)	any two from: 1 same number of each sex ; 2 same / similar, age or five years old ; 3 same, breed / type / AW (of sheep) ; 4 same / similar, mass / size ; 5 do not have, mutation / mutant allele / Huntington's (disease / allele) / neurological disease ;	1
2(b)	any two from: 1 DNA (sequencing), only identifies presence of genes ; 2 RNA (sequencing) identifies, active / switched on / expressed, genes or DNA (sequencing) does not identify, active / switched on / expressed, genes ; 3 more copies of (m)RNA (compared to DNA) ; 4 quantity of RNA indicates level of gene expression ;	2
2(c)(i)	<i>t</i>-test ; comparing means / continuous data / data has a normal distribution / comparing two samples with fewer than 30 values ;	2
2(c)(ii)	there is no difference in the mean counts between the OVT73 and control ;	1

Question	Answer	Marks
2(c)(iii)	any two from: 1 to make a comparison ; 2 to identify if OVT73 (mean) counts was higher / lower (than control) ; 3 use of data from Table 2.1 to support mp2 ;	2
2(d)	max one from contribute 1 two different testing kit used ; 2 commercial kits used, so (presumably) accurate ; 3 statistical tests / analysis were carried out ; max one from improves 4 (test) more sheep ; 5 reanalyse the 10 genes using kit 1, to confirm there was significant difference ; 6 (test) more tissue samples / test the same (tissue) samples again ;	2
2(e)(i)	1.33 ;	1
2(e)(ii)	For anterior striatum, SE (ranges) do not overlap, suggesting there is a <u>significant difference</u> or For cerebellum / motor cortex, SE (ranges) do overlap, so there is not a <u>significant difference</u> ;	1
2(e)(iii)	more urea (in cell), so more transport proteins, more removal of urea ;	1

Question	Answer	Marks
2(f)	any two from: 1 none of the genes coded for enzymes in the urea cycle ; 2 no loss of (striatal) neurones (in OVT73 / transgenic sheep) ; 3 <u>more</u>, urea produced / transport proteins made / gene expression so cells require more energy ;	2